Pre-clinical Operative Dentistry

Prof. Dr. Linz ***Lec: 1***

Operative dentistry

Operative dentistry is the ART & SCIENCE of the diagnosis and treatment of teeth with defects such as caries, discoloration, fractured, malformed which do not require full coverage restorations to return them to the correct form, function and esthetic.

Caries: is an irreversible microbial disease of calcified tissues of the teeth, characterized by demineralization of the inorganic portion and destruction of the organic substance of the tooth which often leads to cavitation.

Classification of carious lesions

G.V. Black who's known as the father of the operative dentistry classified carious lesions into groups according to their locations in permanent teeth

Class I: lesions occur in pits and fissures on facial, lingual and occlusal surfaces of molars and premolars and lingual surface of maxillary anterior teeth (lateral incisors).

Class II: lesions occur on the proximal surfaces of the posterior teeth (molars & premolars)

Class III: lesions occur on the proximal surfaces of the anterior teeth (not on the incisal edge)

Class IV: lesions occur in the proximal surface of the anterior teeth when the incisal angle requires restoration.

Class V: lesions occur on the facial & lingual surfaces in the gingival third of the teeth.

Class VI: lesions are pits or wear defects on the incisal edges of anterior teeth or on cusp tips of posterior teeth.

Note: For abbreviated distal: D, Lingual: L, Mesial: M, Occlusal: O, Incisal: I, Facial: F. dentino-enamel junction: DEJ.

Based on anatomical site caries classified as:

Occlusal (pit & fissure) caries

Smooth surface caries (proximal & cervical caries)

Linear enamel caries which is seen to occur in the region of the neonatal line of the maxillary anterior teeth such as hypocalcemia or truma of birth.

Root surface caries which is more common in older patients as a result of gingival recession, the caries often asymptomatic, rapidly progressed, closer to the pulp and more difficult to restore.

The spread of caries

Bacteria cause caries and caries itself will take the shape of enamel rods, accordingly:

Occlusal caries(pits & fissures) will take the shape of triangle, the base towards the dentin and the tip towards the occlusal surface that is for enamel, whereas for dentin it takes the direction of dentinal tubules, so caries will take the shape of a triangle with the base towards the DEJ and the tip towards the pulp fig1.

In smooth surfaces (proximal surfaces)

In enamel it takes the shape of triangle the base towards the outer surfaces and the tip towards DEJ while in dentin the base of triangle towards the DEJ and the tip towards the pulp fig1.

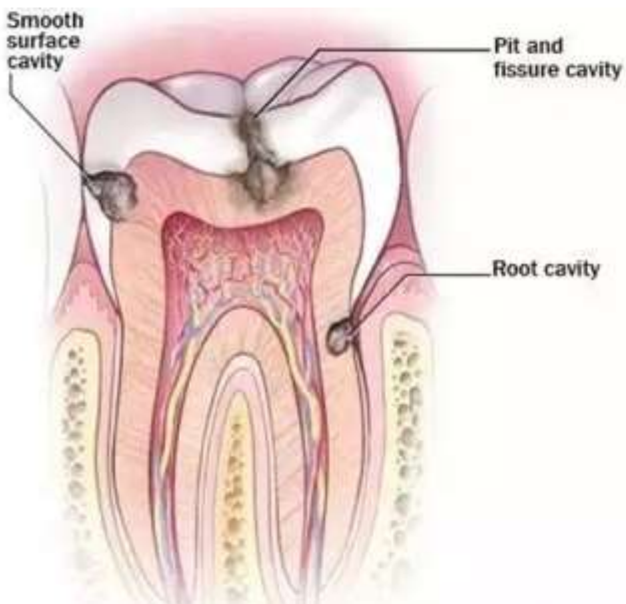

Fig 1: spread of caries in pits and fissures & in smooth surface

Tooth preparation: is a mechanical alteration of the tooth to receive a restorative material which will return the tooth to proper form, function and esthetic.

Objectives of tooth preparation

(1) Conserve as much healthy tooth structure as possible,

(2) remove all defects while simultaneously providing protection of the pulp–dentin complex,

(3) Form the tooth preparation so that, under the forces of mastication, the tooth or the restoration (or both) will not fracture and the restoration will not be displaced,

(4) Allow for the esthetic placement of a restorative material where indicated.

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Factors to Consider Before and During Tooth Preparation

Patient Factors

• Desires • Home care • Risk status • Age • Cooperation • Anesthesia

Lesion/Defect Factors

• Bone support • Occlusion • Severity • Gingival status • Pulpal status • Fracture development

Anatomical Factors

• Enamel rod orientation • Dentin thickness • Pulp location • Coronal contours • Extent of previous restoration

Procedural Factors

• Operator skill • Instrument design • Type of rotary cutting instrument • Ability to isolate

Restorative Material Factors

• Physical properties • Color characteristics • Cost effectiveness.

Tooth Preparation: Terminology

Tooth preparation takes the name of the involved tooth surface(s)—for example, a defect on the occlusal surface is treated with an occlusal preparation.

For brevity in records and communication, the description of a tooth preparation is abbreviated by using the first letter, capitalized, of each tooth surface involved. Examples are as follows:

(1) A simple tooth preparation involving an occlusal surface is an “O”;

(2) A compound preparation involving the mesial and occlusal surfaces is an “MO”; and

(3) A complex preparation involving the mesial, occlusal, distal, and lingual surfaces is an “MODL.”

The process of creating tooth preparation results in the formation of preparation walls or floors fig 2,

An *internal wall*: is a prepared surface that does not extend to the external tooth surface, there are two types of internal walls.

Axial wall: is an internal wall that is oriented parallel to the long axis of the tooth (in cl II cavity preparation). .

Pulpal floor (pulpal wall): is an internal wall that is oriented perpendicular to the long axis of the tooth and is located occlusal to the pulp.

An *external wall:* is a prepared surface that extends to the external tooth surface. Such a wall takes the name of the tooth surface (or aspect) that the wall is adjacent to, as shown in fig 2.

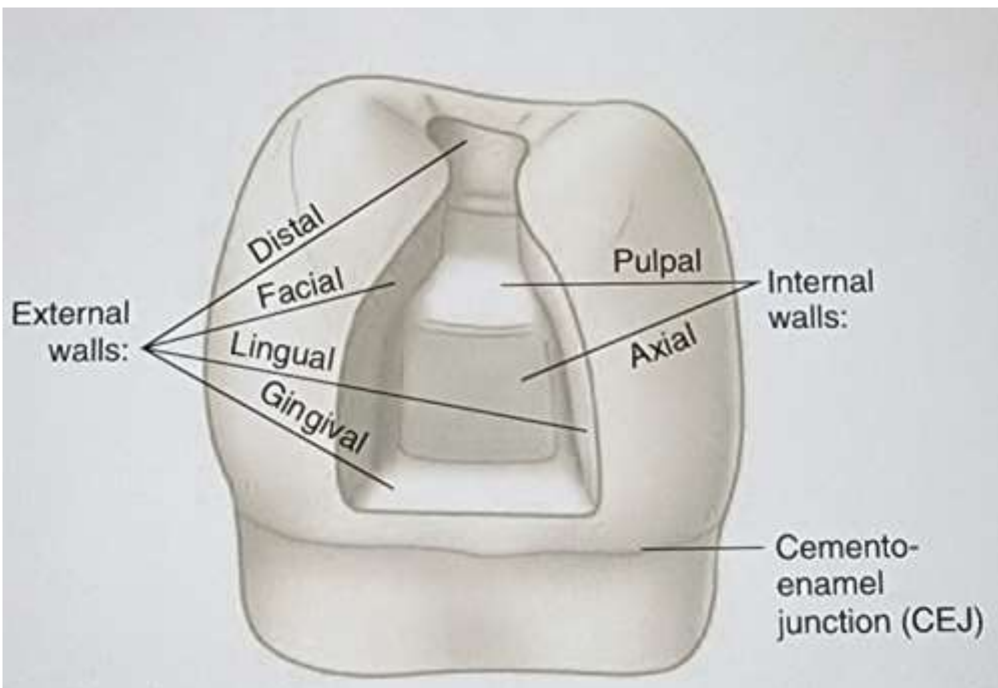

Fig 2: Internal & external tooth preparation walls

Angle: is the junction of two or more prepared surfaces

Line angle: is the junction of two walls in a tooth preparation. Ex. Axio-pulpal line angle fig 3.

Point angle: is the junction of three walls Ex: facio-axio-gingival point angle.

Long axis of the tooth is an imaginary line that pass through the center of the tooth and bisect it to two identical parts.

Cavosurface line angle: is angle formed by the junction of the cavity wall and external tooth surface and are identified by the name of adjacent wall Ex: mesial cavosurface line angle or gingival cavosurface line angle fig 3.

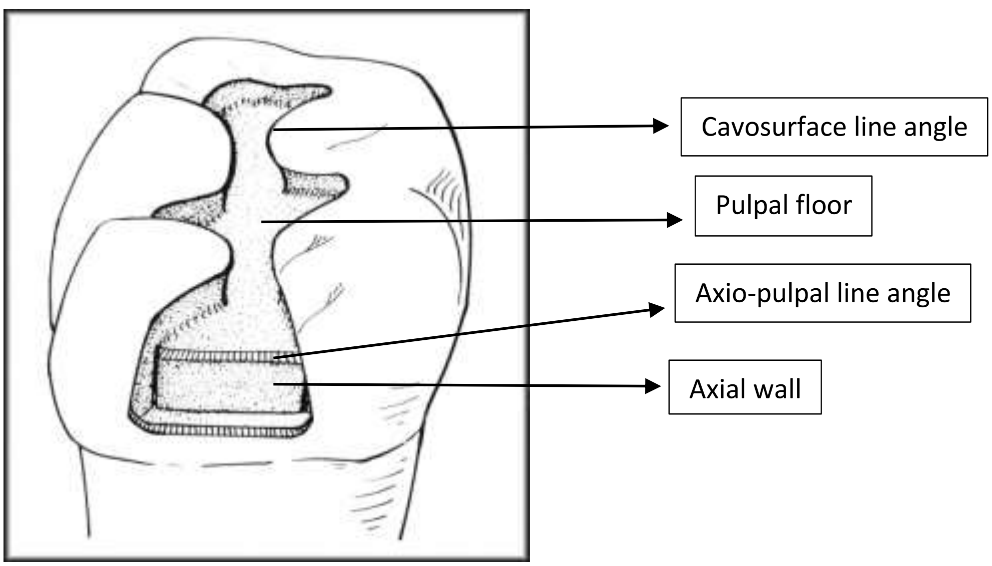

Fig 3: Tooth preparation line angles and walls